

Week 4 Report

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Celebal Excellence Internship

Name: Pranav Deokar

Objective

Understand Azure cloud concepts and build a complete data pipeline using Azure Blob Storage and Azure Data Factory.

Tools Used

Azure Portal, Resource Group, Storage Account, Blob Storage, Azure Data Factory, Azure IAM.

Dataset

Sample Superstore CSV uploaded to the input Blob container.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
1	CA-2016-1	6/16/2016	6/16/2016	Standard	Claire Gutierrez	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-1F	Furniture	Bookcases	Bush Somerset	261.96	2	0	41.9136
2	CA-2016-1	6/16/2016	6/16/2016	Standard	Claire Gutierrez	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-1F	Furniture	Chairs	Honolulu Delux	731.94	3	0	219.582
3	CA-2016-1	6/16/2016	6/16/2016	Standard	Darrin Van	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-1O	Office Supplies	Labels	Self-Adhesive	14.62	2	0	6.8714
4	US-2015-1	10/18/2015	10/18/2015	Standard	Sean O'Donoghue	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-1C	Furniture	Tables	Bretford C	957.5775	5	0.45	-383.031
5	US-2015-1	10/18/2015	10/18/2015	Standard	Sean O'Donoghue	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-1O	Office Supplies	Storage	Eldon Fold	22.368	2	0.2	2.5164
6	CA-2014-1	6/14/2014	6/14/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-1C	Furniture	Furnishings	Eldon Express	48.86	7	0	14.1694
7	CA-2014-1	6/14/2014	6/14/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-1C	Office Supplies	Art	Newell 32	7.28	4	0	1.9656
8	CA-2014-1	6/14/2014	6/14/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-1C	Technology	Phones	Mitel S32C	907.152	6	0.2	90.7152
9	CA-2014-1	6/14/2014	6/14/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-1O	Office Supplies	Binders	DXL Angle	18.504	3	0.2	5.7825
10	CA-2014-1	6/14/2014	6/14/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-1O	Office Supplies	Appliances	Belkin F5C	114.9	5	0	34.47
11	CA-2014-1	6/14/2014	6/14/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-1C	Furniture	Tables	Chromcraft	1706.184	9	0.2	85.3092
12	CA-2014-1	6/14/2014	6/14/2014	Standard	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-1C	Technology	Phones	Konftel 25	911.424	4	0.2	68.3568
13	CA-2017-1	7/16/2017	7/16/2017	Standard	Andrew Ali	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-1O	Office Supplies	Paper	Xerox 196	15.552	3	0.2	5.4432
14	CA-2016-1	6/16/2016	6/16/2016	Standard	Irene Mad	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-1O	Office Supplies	Binders	Fellowes P	407.976	3	0.2	132.5922
15	US-2015-1	11/22/2015	11/26/2015	Standard	Harold Pa	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-1O	Office Supplies	Appliances	Holmes Re	68.81	5	0.8	-123.858
16	US-2015-1	11/22/2015	11/26/2015	Standard	Harold Pa	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-1O	Office Supplies	Binders	Stores Dur	2.544	3	0.8	-3.816
17	CA-2014-1	11/18/2014	11/18/2014	Standard	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-1O	Office Supplies	Storage	Stur-D-Sto	665.88	6	0	13.3176
18	CA-2014-1	5/13/2014	5/15/2014	Standard	Alejandro J	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-1O	Office Supplies	Storage	Fellowes S	55.5	2	0	9.99
19	CA-2014-1	8/27/2014	8/27/2014	Standard	Zuschuss I	Consumer	United States	San Francisco	California	94109	West	OFF-AR-1C	Office Supplies	Art	Newell 34	8.56	2	0	2.4824
20	CA-2014-1	8/27/2014	8/27/2014	Standard	Zuschuss I	Consumer	United States	San Francisco	California	94109	West	TEC-PH-1C	Technology	Phones	Cisco SPA	213.48	3	0.2	16.011
21	CA-2014-1	8/27/2014	8/27/2014	Standard	Zuschuss I	Consumer	United States	San Francisco	California	94109	West	OFF-BI-1O	Office Supplies	Binders	Wilson Jor	22.72	4	0.2	7.384
22	CA-2016-1	12/13/2016	12/13/2016	Standard	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-1C	Office Supplies	Art	Newell 31	19.46	7	0	5.0596
23	CA-2016-1	12/13/2016	12/13/2016	Standard	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-1O	Office Supplies	Appliances	Acco Six-C	60.34	7	0	15.6884
24	US-2017-1	7/16/2017	7/16/2017	Standard	Sandra Fla	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-1F	Furniture	Chairs	Global Del	71.372	2	0.3	-1.0196

Implementation

1. Created Resource Group.
2. Created Storage Account.
3. Created Blob containers (input/output).
4. Uploaded CSV.

5. Created Azure Data Factory.
6. Configured Linked Service.
7. Created Source & Destination Datasets.
8. Added Get Metadata and Copy Data activities.
9. Executed pipeline successfully.
10. Assigned Reader and Contributor roles.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area is divided into two panes. The left pane shows the 'Resource Manager | Resource groups' view with a sidebar containing 'All resources', 'Favorite resources', 'Recent resources', and 'Resource groups'. The right pane shows the 'celebal-week4-rg' resource group overview, which is currently empty, displaying a message: 'No resources match your filters. Try changing or clearing your filters.' Below this, there are buttons for 'Create' and 'Clear filters'.

Below the resource group overview, the breadcrumb navigation shows: Home > Resource Manager | Resource groups > celebal-week4-rg > Marketplace > Storage account. The main heading is 'Create a storage account'.

The 'Create a storage account' form includes the following fields and options:

- Resource group ***: celebal-week4-rg (with a 'Create new' link)
- Instance details**
- Storage account name ***: celebalstorage01
- Region ***: (Asia Pacific) Central India (with a 'Deploy to an Azure Extended Zone' link)
- Primary service ***: Choose preferred storage type (dropdown menu with options: Azure Blob Storage or Azure Data Lake Storage, Azure Files, Other (tables and queues))
- Performance ***: ☐ Premium: Recommended for scenarios that require low latency.
- Redundancy ***: Geo-redundant storage (GRS) (dropdown menu with options: ☒ Make read access to data available in the event of regional unavailability, ☐ Geo priority replication guarantees Blob storage data is geo-replicated within 15 minutes.)

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

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DEFAULT DIRECTORY (PRANAVD...

Home > Resource Manager > Resource groups > celebal-week4-rg > Marketplace > Storage account

Create a storage account

BasicsAdvancedNetworkingData protectionSecurityEncryptionTagsReview + create

[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	celebal-week4-rg
Location	Central India
Storage account name	celebalstorage01
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled

PreviousNextCreate

Give feedback

Microsoft Azure

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Home

celebalstorage01_1783848919676 | Overview

Deployment

Search

DeleteCancelRedeployDownloadRefresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name: celebalstorage01_1783848919676

Subscription: Azure for Students

Resource group: celebal-week4-rg

Start time: 12/07/2026, 15:06:31

Correlation ID: e18e576b-f164-4149-83a5-a96db667856f

Deployment details

Resource	Type	Status	Operation details
celebalstorage01	Microsoft.Storage/storageAccounts	Accepted	Operation details

Give feedback

Tell us about your experience with deployment

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Home

celebalstorage01_1783848919676 | Overview

Deployment

Search

DeleteCancelRedeployDownloadRefresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name: celebalstorage01_1783848919676

Subscription: Azure for Students

Resource group: celebal-week4-rg

Start time: 12/07/2026, 15:06:31

Correlation ID: e18e576b-f164-4149-83a5-a96db667856f

Deployment details

Resource	Type	Status	Operation details
celebalstorage01/default	Microsoft.Storage/storageAccount...	OK	Operation details
celebalstorage01/default	Microsoft.Storage/storageAccount...	OK	Operation details
celebalstorage01	Microsoft.Storage/storageAccounts	OK	Operation details

Next steps

[Go to resource](#)

Give feedback

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Cost Management

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Home > celebalstorage01_1783848919676 | Overview > celebalstorage01 | Containers

input

Container

Search

+ Add Directory ↑ Upload 🔒 Change access level 🔄 Refresh 🗑️ Delete 📄 Copy 📄 Paste 🔄 Rename 🔄 Acquire lease 🔄 Break lease 🛠️ Edit columns

Overview

🔧 Diagnose and solve problems

🔑 Access Control (IAM)

> Settings

input

Authentication method: Access key (Switch to Microsoft Entra user account)

+ Add filter

🔍 Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	📄 Sample - Superstore.csv	7/12/2026, 3:09:49 PM	Hot (Inferred)	Block blob	2.18 MiB	Available	...

Add or remove favorites by pressing **Ctrl+Shift+F**

Home > celebalstorage01

celebalstorage01 | Containers

Storage account

Search

+ Add container ↑ Upload 🔄 Refresh 🗑️ Delete 🔒 Change access level 🔄 Restore containers 🛠️ Edit columns

🔍 Search containers by prefix

Only show active containers

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector	
☐	📁 \$logs	7/12/2026, 3:06:58 PM	Private	Available	-	...
☐	📁 input	7/12/2026, 3:09:13 PM	Private	Available	-	...

Home > celebal-week4-rg > Marketplace > Data Factory

Create Data Factory

🔍 Running final validation...

Basics Git configuration Networking Advanced Tags Review + create

Basics

Subscription	Azure for Students
Resource group	celebal-week4-rg
Name	celebaladf01
Region	Central India
Version	V2

Networking

Connect via	Public endpoint
-------------	-----------------

Previous Next Create

🔗 Give feedback

Microsoft Azure

Search resources, services, and docs (G+)

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DEFAULT DIRECTORY (PRANAVD...)

Home

Microsoft.DataFactory-20260712151126 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.DataFactory-20260712151126
Subscription : Azure for Students
Resource group : celebal-week4-rg

Start time : 12/07/2026, 15:12:59
Correlation ID : 414ec9e9-7531-4de1-87b1-64886bea5cbb

Deployment details

Resource	Type	Status	Operation details
celebaladf01	Data factory (V2)	OK	Operation details

Next steps

Go to resource

Give feedback

Tell us about your experience with deployment

Deployment succeeded

Deployment 'Microsoft.DataFactory-20260712151126' to resource group 'celebal-week4-rg' was successful.

Pin to dashbo... Go to resource gr...

Cost management

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Microsoft Azure

Data Factory ▶ celebaladf01

Search

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Plan ahead with Fabric

Azure Data Factory allows you to configure a Git repository with either Azure DevOps or GitHub. Git is a version control system that allows for easier change tracking and collaboration. [Learn more](#)

Set up code repository

Data factory

celebaladf01

New

Ingest

Copy data at scale once or on a schedule.

Orchestrate

Code-free data pipelines.

Transform data

Transform your data using data flows.

Configure SSIS

Manage & run your SSIS packages in the cloud.

Recent resources

No items to show.

New linked service

 Azure Blob Storage [Learn more](#) 

Name *

LS_BlobStorage

Description

Connect via integration runtime * ⓘ

 AutoResolveIntegrationRuntime

Authentication type

Account key

Connection string

Azure Key V

Account key

Account selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription ⓘ

Azure for Students (39a1aa9d-bb0b-449d-9f62-583a392815ef)

Storage account name *

celebalstorage01

Additional connection properties

Create

Back



Test connection

Cancel

Microsoft Azure | Data Factory | celebalad01

Search

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Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

» Data Factory | Validate all | Publish all

Migrate to Fabric (Preview)

General

Connector upgrade advis...

Factory settings

Connections

Linked services

Integration runtimes

Microsoft Purview

ADF in Microsoft Fabric

Source control

Git configuration

ARM template

Author

Triggers

Global parameters

Data flow libraries

Security

Credentials

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations : Any

Showing 1 - 1 of 1 items

Name	Type	Related	Annotations
LS_BlobStorage	Azure Blob Storage	0	

Set properties

Name

DS_Source

Linked service *

LS_BlobStorage

File path

input

Directory

Sample - Superstore.csv

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

Microsoft Azure | Data Factory | celebalad01

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Factory Resources

- Pipelines: 0
- Change Data Capture (preview): 0
- Datasets: 1
 - DS_Source
- Data flows: 0
- Power Query: 0

DS_Source

DelimitedText
DS_Source

Properties

General

Name *
DS_Source

Description

Annotations
+ New

Cancel

Connection

Linked service *
LS_BlobStorage

File path *
Input / Directory / Sample - Superstore.csv

Compression type
No compression

Column delimiter
Comma (,)

Row delimiter
Default (\r\n, or \n)

Encoding
Default(UTF-8)

Microsoft Azure | Data Factory | celebalad01

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Activities

- Move and transform
 - Copy data
 - Data flow
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
 - Databricks
- General
 - Append variable
 - Delete
 - Execute Pipeline
 - Execute SSIS package

PL_Copy_CSV

Get Metadata

Copy data

Copy data1

Properties

General

Name *
PL_Copy_CSV

Description

Annotations
+ New

Cancel

Source

Source dataset *
DS_Source

File path type
☒ File path in dataset ☐ Prefix ☐ Wildcard file path ☐ List of files

Start time (UTC)

End time (UTC)

Microsoft Azure | Data Factory | celebalad01

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Activities

- Move and transform
 - Copy data
 - Data flow
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
 - Databricks
- General
 - Append variable
 - Delete
 - Execute Pipeline
 - Execute SSIS package

PL_Copy_CSV

Copy data

Properties

General

Name *
PL_Copy_CSV

Description

Annotations
+ New

Cancel

Sink

Sink dataset *
DS_Destination

Copy behavior
Select...

Max concurrent connections

Block size (MB)

Metadata
+ New

Quote all text
☒

File extension
.txt

Max rows per file

Microsoft Azure | Data Factory | celebalad01

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

DS_Source | DS_Destination | PL_Copy_CSV

Activities

- Move and transform
 - Copy data
 - Data flow
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
 - Databricks
- General
 - Append variable
 - Delete
 - Execute Pipeline
 - Execute SSIS package

Parameters Variables Settings Output

Pipeline run ID: 3d897cd9-8fe1-4a11-8859-6b06d6d71ea1 | Pipeline status: Succeeded

View debug run consumption | Monitor in Azure Metrics | Export to CSV

Showing 1 - 2 of 2 items

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	7/12/2026, 3:38:34 PM	15s	AutoResolveIntegrationRuntime
Get Metadata1	Succeeded	Get Metadata	7/12/2026, 3:38:20 PM	13s	AutoResolveIntegrationRuntime

Properties

General Related

Name: PL_Copy_CSV

Description:

Annotations

+ New

Cancel

Microsoft Azure | Superstore_Copy.csv

Home | Microsoft.DataFactory-20260712151126 | Overview

output

Containers

Overview

- Diagnose and solve problems
- Access Control (IAM)
- Settings

Authentication methods

Add filter

Search blobs by

Showing all 1 items

Name

Superstore_Copy.csv

Overview Versions Snapshots Edit Generate SAS

Properties

URL: https://celebalstorage01.blob.core.windows.net/superstore/

LAST MODIFIED: 12/07/2026, 3:38:45 pm

CREATION TIME: 12/07/2026, 3:38:45 pm

VERSION ID: -

TYPE: Block blob

SIZE: 2.18 MiB

ACCESS TIER: Hot (Inferred)

ACCESS TIER LAST MODIFIED: N/A

ARCHIVE STATUS: -

REHYDRATE PRIORITY: -

SERVER ENCRYPTED: true

ETAG: 0x8DEDFD8F087199

VERSION-LEVEL IMMUTABILITY POLICY: Disabled

CACHE-CONTROL:

CONTENT-TYPE: application/octet-stream

CONTENT-MD5: swZpBenTUD0U9Rz2/IMAg...

CONTENT-ENCODING:

Microsoft Azure | celebalstorage01 | Access Control (IAM)

Home | Microsoft.DataFactory-20260712151126 | Overview | celebal-week4-rg | celebalstorage01

Storage account

Search

Diagnose and solve problems

Access Control (IAM)

- Data migration
- Events
- Storage browser
- Storage Mover
- Partner solutions
- Resource visualizer
- Data storage
 - Containers
 - Classic file shares
 - Queues
 - Tables
- Security + networking
- Data management
- Settings

Download role assignments | Edit columns | Refresh | Delete | Feedback

Search by name or email

Type: All | Role: All | Scope: All scopes | Group by: Role

All (4) | Job function roles (1) | Privileged administrator roles (3)

Name	Type	Role	Scope	Condition
Owner (2)				
pranav deokar 8977285-b0b9-42da-b221-4...	User	Owner	Subscription (Inherited)	None
pranav deokar 8977285-b0b9-42da-b221-4...	User	Owner	Subscription (Inherited)	None
Contributor (1)				
celebalad01 1c372264-075d-4617-aac5-...	Managed identity	Contributor	This resource	None
Reader (1)				
celebalad01 1c372264-075d-4617-aac5-...	Managed identity	Reader	This resource	None

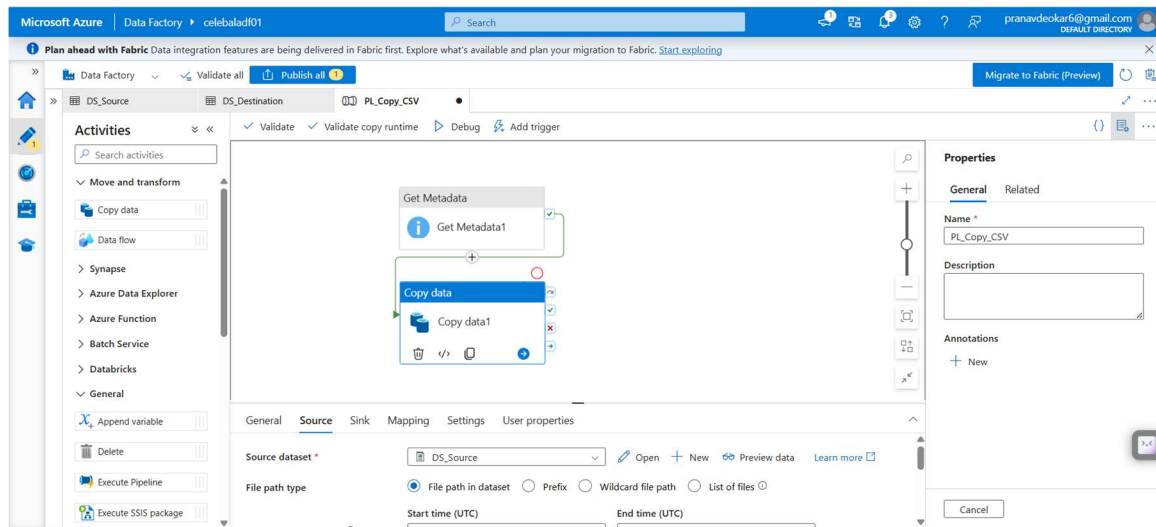
Showing 1 - 4 of 4 results.

Pipeline Workflow

Sample-Superstore.csv → Input Blob → Linked Service → Source Dataset → Get Metadata → Copy Data → Output Blob → Superstore_Copy.csv

Result

Pipeline executed successfully, metadata validated and CSV copied successfully to the output container.



Learning Outcomes

- Azure Resource Management
- Blob Storage
- Azure Data Factory
- Linked Services
- Datasets
- IAM

Conclusion

Successfully implemented an end-to-end Azure data pipeline using Azure Blob Storage and Azure Data Factory.